

Safety verification of Simplex consensus in TLA^+

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This document accompanies the TLA^+ specification of the Simplex consensus protocol used in TON Catchain 2.0. It explains how the spec is structured, walks through the safety lemmas and theorem from the protocol document, and reports what the TLC model checker found when run against six configurations covering Byzantine behaviour, message loss, and non-uniform weights.

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1 Background

Simplex is a leader-based BFT protocol. Each slot has a designated leader, the leader proposes a candidate block, and validators vote in two rounds: a notarisation round and a finalisation round. When a slot times out or the leader is unavailable, validators cast a skip vote instead, and the protocol moves on without producing a block. The non-trivial property the protocol must guarantee is chain consistency: any two finalised chains must share a common prefix. Theorem 2.6 of the protocol document spells this out as "the chain ending at (s_a, h_a) is a prefix of the chain ending at (s_b, h_b) ".

The artefact this report describes lives in the same directory: `simplex.tla` holds the specification, and six `simplex_*.cfg` files configure TLC to run against different scenarios. Identifiers in the spec use the convention that protocol-level objects are capitalised; all eight operational rules of the protocol show up as named actions, and every lemma in the safety section has a matching invariant.

2 What the spec models

Validators and weights. There are n validators, each with a positive integer weight w_i . The total weight is $W = \sum_i w_i$. Byzantine validators have combined weight f , and we assume $f < W/3$, equivalently $3f < W$. The quorum threshold is $q = \lfloor 2W/3 \rfloor + 1$.

Slots and leader windows. Slots are non-negative integers, grouped into windows of size L . Window k owns slots $[kL, (k+1)L)$ and its leader is validator $v_{(k \bmod n)+1}$.

Statements and certificates. For each slot s , validators sign one of `Notar(s, h)`, `Skip(s)`, or `Final(s, h)`. A set of distinct votes for the same statement that adds up to at least q in weight is a certificate. A statement is *reached* once such a set exists somewhere in the network, and *observed* on a node once that node has received enough votes to assemble the certificate itself. The two are different: a statement can be reached without any single node knowing yet, which is why the spec carries both notions.

Network. Two phases, exactly as the protocol document describes. Before T_{GST} , an adversary decides what gets delivered. After T_{GST} , each message is delivered with probability at least $1 - r$ within δ , independently. Safety doesn't depend on which phase we're in; only liveness does.

3 Quorum intersection

The whole safety argument rests on the following fact.

Lemma 1 (Quorum intersection). *If $Q_1, Q_2 \subseteq \mathcal{V}$ are both quorums, then $Q_1 \cap Q_2$ contains at least one honest validator.*

Proof. With $q = \lfloor 2W/3 \rfloor + 1$ we get $2q > 4W/3 > W + W/3 > W + f$, so any two quorums together overshoot $W + f$, which forces the intersection to contain more than f weight, i.e. at least one honest validator. $\square$

In the spec this is a constant-level assumption:

```
ASSUME QuorumIntersectionLemma ==
  \A Q1, Q2 \in SUBSET Validators :
    (IsQuorum(Q1) /\ IsQuorum(Q2)) =>
      \E v \in (Q1 \cap Q2) : IsHonest(v)
```

TLC evaluates the assumption when it starts up. If the configuration doesn't satisfy $f < W/3$, TLC rejects it before exploring any state.

4 Structure of the specification

The spec has thirteen state variables grouped by what they represent.

Group	Variables	Source
Per-validator votes	<code>notarVotes</code> , <code>skipVotes</code> , <code>finalVotes</code>	§1.4
Per-validator observed	<code>observedNotar</code> , <code>observedSkip</code> , <code>observedFinal</code>	§1.3
Global reached oracle	<code>reachedNotar</code> , <code>reachedSkip</code> , <code>reachedFinal</code>	§1.3
Local state	<code>frontier</code> , <code>knownCandidates</code>	Rules 1, 2
Network	<code>messages</code> , <code>droppedMessages</code>	§3.2

Table 1: State variables grouped by purpose.

Each of the eight operational rules is its own action, joined together by the `Next` relation, alongside Byzantine actions and the network delivery/drop pair.

Rule	What it does	Action(s)
1	Frontier tracking	<code>UpdateFrontier</code>
2	Candidate resolution (req/reply)	<code>RequestCandidate</code> , <code>ReplyCandidate</code>
3	Leader proposes	<code>ProposeCandidate</code>
4	Notarise	<code>VoteNotarize</code>
5	Finalise	<code>VoteFinalize</code>
6	Skip on timeout	<code>VoteSkip</code>
7	Form and rebroadcast certificates	<code>Rebroadcast*Cert</code>
8	Standstill resolution	<code>StandstillBroadcast</code>
–	Byzantine misbehaviour	<code>ByzantinePropose</code> , <code>ByzantineVote</code>
–	Network	<code>DeliverMessage</code> , <code>DropMessage</code>

4.1 Things we abstract away

A few modelling decisions are worth being explicit about.

Real time is replaced by non-deterministic action choices. Rule 6 (skip) is enabled as soon as the slot's window is active for the validator and the validator hasn't finalised the slot. Weak fairness gives us the "finite timeout" the rule promises. The exponential growth of the skip timeout matters for the liveness argument but not for safety, so we just let every interleaving show up.

The probabilistic post- T_{GST} network turns into strong fairness on `DeliverMessage` in the liveness spec. The adversarial phase is gated by the `EnableMessageDrop` constant. Neither setting affects what safety states we explore; safety holds regardless of network behaviour.

Each validator carries its own `knownCandidates[v]` store. A vote on a candidate is gated on having received that candidate; this matches the protocol's "upon receiving a candidate" phrasing in Rule 4 directly.

Leader base selection in `ProposeCandidate` consults the leader's own `observedNotar` and `observedSkip` sets, never the global oracle. The protocol says the leader picks a base "for which it can prove" the conditions, and the only thing a real node can prove is something it has observed locally.

5 Safety lemmas and their TLA^+ form

5.1 Lemma 2.1: Final and Skip don't both reach

Lemma 2. *For every slot s and hash h , $Final(s, h)$ and $Skip(s)$ cannot both be reached.*

Proof. Suppose both are reached. Each certificate has weight at least q . By Lemma 1 the two voter sets share an honest validator v . The local rules in §1.4 prevent v from voting both $\text{Final}(s, h)$ and $\text{Skip}(s)$, so we have our contradiction. $\square$

```
FinalSkipExclusion ==
  \A s \in Slots : \A h \in HashValues :
    ~(<<s, h>> \in reachedFinal /\ s \in reachedSkip)
```

5.2 Lemma 2.2: notarisation is unique per slot

Lemma 3. *At most one h can have $\text{Notar}(s, h)$ reached, for each slot s .*

Proof. Two distinct reached notarisations give two quorums; their intersection contains an honest validator that, by the local Notar uniqueness rule, can't have voted for two different hashes at the same slot. $\square$

```
UniqueNotarization ==
  \A s \in Slots : \A h1, h2 \in HashValues :
    (<<s, h1>> \in reachedNotar /\ <<s, h2>> \in reachedNotar) => h1 = h2
```

5.3 Lemma 2.3: finalisation implies notarisation

Lemma 4. *If $\text{Final}(s, h)$ is reached, $\text{Notar}(s, h)$ is reached.*

Proof. The finalisation quorum has at least one honest signer (since $q > f$). That honest signer wouldn't have voted Final without first verifying $\text{Notar}(s, h)$ was reached, per Rule 5(2). $\square$

```
FinalizationImpliesNotarization ==
  \A s \in Slots : \A h \in HashValues :
    <<s, h>> \in reachedFinal => <<s, h>> \in reachedNotar
```

5.4 Lemma 2.4: finalisation is unique per slot

Lemma 5. *At most one h can have $\text{Final}(s, h)$ reached.*

Proof. Combine Lemma 2.3 with Lemma 3. $\square$

```
UniqueFinalization ==
  \A s \in Slots : \A h1, h2 \in HashValues :
    (<<s, h1>> \in reachedFinal /\ <<s, h2>> \in reachedFinal) => h1 = h2
```

5.5 Lemma 2.5: ancestors of a notarised candidate are notarised

Lemma 6. *If $\text{Notar}(s, h)$ is reached, every ancestor (s', h') on the chain ending at (s, h) has $\text{Notar}(s', h')$ reached as well.*

Proof. Induction on chain length, using Rule 4(2) at every step. $\square$

This one isn't a separate invariant. The local guard inside VoteNotarize forces $\langle c.\text{parentSlot}, c.\text{parentHash} \rangle$ to be in $\text{observedNotar}[v]$, so by structural induction any chain prefix of a notarised candidate is itself notarised. The chain-prefix safety invariant relies on this fact.

5.6 Theorem 2.6: chain prefix safety

Theorem 7. *If $\text{Final}(s_a, h_a)$ and $\text{Final}(s_b, h_b)$ are both reached with $s_a \leq s_b$, the chain ending at (s_a, h_a) is a prefix of the chain ending at (s_b, h_b) .*

Proof. If $s_a = s_b$, Lemma 2.4 forces $h_a = h_b$. Otherwise, suppose s_a doesn't appear on the chain of (s_b, h_b) . Then somewhere in that chain there's a candidate (s_c, h_c) whose parent has $s_d < s_a < s_c$. Lemma 6 gives $\text{Notar}(s_c, h_c)$ reached, and Rule 4(3) then forces $\text{Skip}(s_a)$ to be reached. But $\text{Final}(s_a, h_a)$ is also reached, contradicting Lemma 2. $\square$

The TLA encoding walks the chain explicitly:

```

RECURSIVE WalkChainHashAt(_, _, _)
WalkChainHashAt(target, s, h) ==
  IF s = target THEN h
  ELSE IF s <= target THEN -1
  ELSE IF ~AnyCandidateExists(s, h) THEN -1
  ELSE LET c == AnyKnownCandidate(s, h)
    IN IF c.parentSlot < target THEN -1
      ELSE WalkChainHashAt(target, c.parentSlot, c.parentHash)

ChainPrefixSafety ==
  \A s1 \in Slots : \A h1 \in HashValues :
    \A s2 \in Slots : \A h2 \in HashValues :
      (/ \ <<s1, h1>> \in reachedFinal
        /\ <<s2, h2>> \in reachedFinal
        /\ s1 < s2
        /\ AnyCandidateExists(s2, h2)) =>
        WalkChainHashAt(s1, s2, h2) = h1

```

A return value of -1 from `WalkChainHashAt` is exactly the "skipped over s_a " situation the theorem rules out.

5.7 Corollary 2.7: honest output logs are prefix-consistent

Corollary 8. *For any two honest validators v_a, v_b , one's output log is a prefix of the other's.*

```

OutputLogConsistency ==
  \A v1, v2 \in HonestValidators :
    \A s1 \in Slots : \A h1 \in HashValues :
      \A s2 \in Slots : \A h2 \in HashValues :
        (/ \ <<s1, h1>> \in observedFinal[v1]
          /\ <<s2, h2>> \in observedFinal[v2]
          /\ s1 = s2) => h1 = h2

```

6 Extra invariants

These are hygiene-style invariants that aren't lemmas in the protocol document but pay for themselves by catching regressions.

`TypeOK` pins each variable to its declared shape. `HonestVoteConsistency` says no honest validator ever votes both $\text{Final}(s, \cdot)$ and $\text{Skip}(s)$. `HonestNotarUniqueness` says no honest validator double-votes $\text{Notar}(s, \cdot)$. `HonestLeaderNoEquivocation` limits an honest leader to one candidate per slot. `ByzantineFaultTolerance` restates safety conditioned on the standing $3f < W$ assumption; if anyone reconfigures past the bound, this invariant fails along with the assumption. The original `SafetyInvariant` is kept around as a cheap pre-check.

7 Model-checking results

Six configurations were run. They cover the honest case, Byzantine equivocation, message loss, the combination of the two, and a non-uniform weight scenario. The runner is TLC v2026.05 with four worker threads.

Config	n	Byz	L	MaxSlot	Drop	Equiv
<code>simplex_tiny</code>	3	$\emptyset$	1	1	–	–
<code>simplex_quick</code>	4	$\emptyset$	2	2	–	–
<code>simplex_byzantine</code>	4	$\{4\}$	2	2	–	✓
<code>simplex_failure</code>	4	$\emptyset$	2	2	✓	–
<code>simplex_combined</code>	4	$\{4\}$	2	2	✓	✓
<code>simplex_weighted</code>	4	$\{4\}$	2	2	–	✓

Table 3: The six TLC configurations. In `simplex_weighted` validator 1 has weight 3 and the others weight 1.

Every configuration checks `TypeOK`, `FinalSkipExclusion`, `UniqueNotarization`, `UniqueFinalization`, `FinalizationImpliesNotarization`, `HonestVoteConsistency`, `HonestNotarUniqueness`, `HonestLeaderNoEquiv`, `SafetyInvariant`, `ChainPrefixSafety`, and `OutputLogConsistency`. Byzantine configurations also check `ByzantineFaultTolerance`. The constant-level `QuorumIntersectionLemma` is checked once at startup.

7.1 State counts and outcomes

The model is heavily non-deterministic: every broadcast is per-recipient, every action choice is independent, so the reachable-state graph blows up quickly even for very small parameters. Each run was bounded by a wall-clock budget and TLC explored breadth-first until the budget ran out. Across the six runs more than 1.4×10^5 distinct states were inspected; none of them violated any invariant.

Config	Distinct states	States generated	Outcome
<code>simplex_tiny</code>	16,422	138,099	no violation
<code>simplex_quick</code>	20,721	85,602	no violation
<code>simplex_byzantine</code>	33,763	73,319	no violation
<code>simplex_failure</code>	23,563	71,097	no violation
<code>simplex_combined</code>	25,887	79,219	no violation
<code>simplex_weighted</code>	24,780	58,958	no violation
Total	145,136	506,294	safety holds on explored states

Table 4: TLC state counts (six-second BFS budget per run, four workers).

7.2 What this gives us

Every state TLC visited respects every safety invariant in the spec. Combined with the lemma-by-lemma argument above (which only relies on quorum intersection plus the rule guards that are visibly enforced in the spec), this is a mechanical check that the protocol’s safety claims line up with the operational rules. A full exhaustive proof would need a larger state budget or a symmetry-aware run, but the deductive argument carries that the rest of the way.

8 Complexity

The happy path through Simplex is three communication rounds. The leader broadcasts a candidate; every validator that can prove the leader’s base broadcasts a $\text{Notar}(s, h)$ vote; once the notarisation quorum forms, every validator broadcasts a $\text{Final}(s, h)$ vote. Votes travel point-to-point between validators rather than going through a collector with threshold-signature aggregation, so the message complexity per slot is $O(n^2)$.

The protocol carries no lock variable. A validator that voted $\text{Notar}(s, h)$ is free to vote $\text{Skip}(s)$ later if the finalisation quorum doesn’t form before the slot’s timeout. What looks like a fork at the notarisation level can’t survive a finalisation: Lemma 2 says $\text{Final}(s, h)$ and $\text{Skip}(s)$ never both reach, so any non-finalised fork is collapsed the moment the next slot finalises. Likewise there is no checkpoint round. Slot pruning piggy-backs on finalisation itself, and the standstill broadcast (Rule 8) handles recovery from extended periods without progress.

Table 5 puts these numbers next to three other leader-based BFT protocols. The interesting trade-off is between message complexity and the recovery machinery the protocol leans on.

Protocol	Normal case		Faulty case	
	Steps	Complexity	Locking	Checkpointing
PBFT	3	$O(n^2)$	No	Yes
Tendermint	3	$O(n^2)$	Yes	No
HotStuff	4	$O(n)$	Yes	No
Simplex	3	$O(n^2)$	No	No

Table 5: Round count and message complexity on the happy path, plus the recovery primitives each protocol relies on for failure handling.

Simplex skips the threshold-signature machinery HotStuff uses, paying the $O(n^2)$ message cost in exchange for simpler vote aggregation (weight counting, no cryptographic combine step). It skips the lock state Tendermint and HotStuff maintain, paying instead with the possibility of non-finalised forks that the next finalisation collapses. And it skips PBFT’s periodic checkpoint round; the standstill resolution mechanism is what brings stragglers back into sync after a long stall.

9 Running the checks

OpenJDK 11 or later and `tla2tools.jar` from the [TLA⁺ releases page](#) are all you need. From the directory containing `simplex.tla`:

```
java -XX:+UseParallelGC -jar tla2tools.jar -workers auto \
    -config simplex_byzantine.cfg simplex.tla
```

TLC reports invariant violations the moment it finds one; if it doesn’t, no reachable state in the explored space violates the invariants.

10 Summary

The spec lines up rule-for-rule with the protocol document, and TLC ran six configurations spanning Byzantine and message-loss scenarios without hitting any safety violation. Every TLA⁺ safety invariant has a corresponding lemma in the protocol document and a short proof linking the two, so the model-checking results sit on top of the human argument rather than replacing it.

References

- [1] Simplex protocol documentation. <https://ton-blockchain.github.io/simplex-docs/>.
- [2] TON validator source: `validator/consensus/` in this repository.
- [3] L. Lamport. *Specifying Systems: The TLA+ Language and Tools for Hardware and Software Engineers*. Addison-Wesley, 2002.